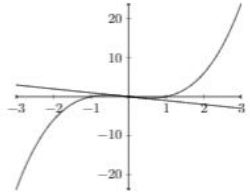
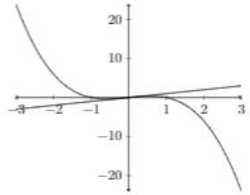
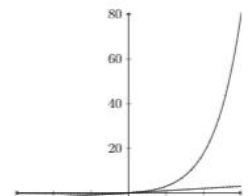


Week 8 Solve With Us

Mathematics 1

Q1. Match the given functions in Column A with the equations of their tangents at the origin $(0, 0)$ in column B and the plotted graphs and the tangents in Column C, given in Table M2W2G1.

	Function (Column A)		It's tangent at $(0,0)$ (Column B)		Graph (Column C)
i)	$f(x) = x3^x$	a)	$y = -x$	1)	
ii)	$f(x) = x(x-1)(x+1)$	b)	$y = x$	2)	
iii)	$f(x) = -x(x-1)(x+1)$	c)	$y = x$	3)	

☐ Option 1: ii) $\rightarrow$ a) $\rightarrow$ 1.

☐ Option 2: i) $\rightarrow$ b) $\rightarrow$ 3.

☐ Option 3: iii) $\rightarrow$ b) $\rightarrow$ 1.

☐ Option 4: iii) $\rightarrow$ c) $\rightarrow$ 2.

☐ Option 5: i) $\rightarrow$ a) $\rightarrow$ 1.

- ☐ **Option 1:** ii) $\rightarrow$ a) $\rightarrow$ 1.
- ☐ **Option 2:** i) $\rightarrow$ b) $\rightarrow$ 3.
- ☐ **Option 3:** iii) $\rightarrow$ b) $\rightarrow$ 1.
- ☐ **Option 4:** iii) $\rightarrow$ c) $\rightarrow$ 2.
- ☐ **Option 5:** i) $\rightarrow$ a) $\rightarrow$ 1.

Solution

i) Given $f(x) = x3^x \implies f'(x) = 3^x + x3^x \ln 3$. So, $f(0) = 0$ and $f'(0) = 1$.
Hence the equation of the tangent at the origin is

$$y - 0 = 1.(x - 0) \implies y = x.$$

In Column C, figure 3 has the line $y = x$ and exponential graph.

Hence i) $\rightarrow$ b) $\rightarrow$ 3).

ii) Given $f(x) = x(x-1)(x+1) = x^3 - x \implies f'(x) = 3x^2 - 1$.

So, $f(0) = 0$ and $f'(0) = -1$

Hence the equation of the tangent at the origin is

$$y - 0 = -1(x - 0) \implies y = -x.$$

In Column C, figure 1 has the line $y = -x$.

Hence ii) $\rightarrow$ a) $\rightarrow$ 1).

iii) Given $f(x) = -x(x-1)(x+1) = -x^3 + x \implies f'(x) = -3x^2 + 1$.

So, $f(0) = 0$ and $f'(0) = 1$

Hence the equation of the tangent at the origin is

$$y - 0 = 1 \times (x - 0) \implies y = x$$

In Column C, figure 2 has the line $y = x$.

Hence iii) $\rightarrow$ c) $\rightarrow$ 2).

L'Hospital's rule (*loh-pee-TAHL*)

L'Hôpital's rule states that for functions f and g which are **differentiable** on an open **interval** I except possibly at a point c contained in I , if $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$ or $\pm \infty$, and $g'(x) \neq 0$ for all x in I with $x \neq c$, and $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$ exists, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}.$$

Q2

Choose the set of correct options considering the function given below:

$$f(y) = \begin{cases} \frac{\sin y}{y} & \text{if } y \neq 0, \\ 1 & \text{if } y = 0 \end{cases}$$

- ☐ Option 1: $f(y)$ is not continuous at $y = 0$.
- ☐ Option 2: $f(y)$ is continuous at $x = 0$.
- ☐ Option 3: $f(y)$ is not differentiable at $y = 0$.
- ☐ Option 4: $f(y)$ is differentiable at $y = 0$.
- ☐ Option 5: The derivative of $f(y)$ at $y = 0$ (if exists) is 0.
- ☐ Option 6: The derivative of $f(y)$ at $y = 0$ (if exists) is 1.

- ☐ Option 1: $f(y)$ is not continuous at $y = 0$.
- ☐ **Option 2:** $f(y)$ is continuous at $x = 0$.
- ☐ Option 3: $f(y)$ is not differentiable at $y = 0$.
- ☐ **Option 4:** $f(y)$ is differentiable at $y = 0$.
- ☐ **Option 5:** The derivative of $f(y)$ at $y = 0$ (if exists) is 0.
- ☐ Option 6: The derivative of $f(y)$ at $y = 0$ (if exists) is 1.

Solution:

We know that $\lim_{y \rightarrow 0} \frac{\sin y}{y} = 1 = f(0)$. So $f(y)$ is continuous at $y = 0$.

Hence option 2 is true.

$$\text{Now, } \lim_{c \rightarrow 0} \frac{f(0+c)-f(0)}{c} = \lim_{c \rightarrow 0} \frac{f(c)-1}{c} = \lim_{c \rightarrow 0} \frac{\frac{\sin c}{c}-1}{c} = \lim_{c \rightarrow 0} \frac{\sin c - c}{c^2} = \lim_{c \rightarrow 0} \frac{\cos c - 1}{2c} = \lim_{c \rightarrow 0} \frac{-\sin c}{2} = 0$$

(using L'Hopital's rule twice).

Hence the derivative of $f(y)$ at $y = 0$ is 0.

So options 4 and 5 are true.

Q3

Let g be a polynomial of degree 5, which is given by

$$g(y) = b_5y^5 + b_4y^4 + b_3y^3 + b_2y^2 + b_1y + b_0.$$

Let $g'(c)$ denote the derivative of g at $y = c$. Choose the set of correct options.

- ☐ Option 1: $b_1 = g'(0)$
- ☐ Option 2: $5b_5 + 3b_3 = \frac{1}{2}(g'(1) + g'(-1) - 2g'(0))$
- ☐ Option 3: $4b_4 + 2b_2 = \frac{1}{2}(g'(1) - g'(-1))$
- ☐ Option 4: None of the above.

Solution

- ☐ **Option 1:** $b_1 = g'(0)$
- ☐ **Option 2:** $5b_5 + 3b_3 = \frac{1}{2}(g'(1) + g'(-1) - 2g'(0))$
- ☐ **Option 3:** $4b_4 + 2b_2 = \frac{1}{2}(g'(1) - g'(-1))$
- ☐ **Option 4:** None of the above.

Solution:

Given $g(y) = b_5y^5 + b_4y^4 + b_3y^3 + b_2y^2 + b_1y + b_0 \implies g'(y) = 5b_5y^4 + 4b_4y^3 + 3b_3y^2 + 2b_2y + b_1$

So $g'(0) = b_1$, so option 1 is correct. $g'(1) = 5b_5 + 4b_4 + 3b_3 + 2b_2 + b_1$, and $g'(-1) = 5b_5 - 4b_4 + 3b_3 - 2b_2 + b_1$

Hence $5b_5 + 3b_3 = \frac{1}{2}(g'(1) + g'(-1) - g'(0))$ and $4b_4 + 2b_2 = \frac{1}{2}(g'(1) - g'(-1))$. So options 2, 3 are correct.

Q4

The distance (in meters) traveled by a car after t minutes is given by the function $c(t) = f(t^3 + t^2 + t + 1)$, where f is a differentiable function with domain $\mathbb{R}$. Find the instantaneous speed of the car after 5 min ($t = 5$), where $f'(156) = 1$.

The distance (in meters) traveled by a car after t minutes is given by the function $c(t) = f(t^3 + t^2 + t + 1)$, where f is a differentiable function with domain $\mathbb{R}$. Find the instantaneous speed of the car after 5 min ($t = 5$), where $f'(156) = 1$.

Solution:

The instantaneous speed of the car after t min $= f'(t) = f'(t^3 + t^2 + t + 1)(3t^2 + 2t + 2)$.
(use derivative property of composition of two functions)

So the instantaneous speed of the car after 5 min $= f'(156) \cdot 86 = 86$

Q5

Consider the following function

$$p : \mathbb{R} \rightarrow \mathbb{R}$$

$$p(x) = |x(x - 7)(x - 8)|.$$

If m is slope of the tangent of the function $p(x)$ at point $x = \frac{1}{2}$, find the value $m - \frac{3}{4}$.

Consider the following function

$$p : \mathbb{R} \rightarrow \mathbb{R}$$

$$p(x) = |x(x - 7)(x - 8)|.$$

If m is slope of the tangent of the function $p(x)$ at point $x = \frac{1}{2}$, find the value $m - \frac{3}{4}$.

Solution:

Observe that $p(x) = x(x - 7)(x - 8) = x^3 - 15x^2 + 56x$ if $0 \leq x < 7$.

So $p'(x) = 3x^2 - 30x + 56 \implies p'(\frac{1}{2}) = \frac{3}{4} - 15 + 56 = \frac{3}{4} + 41$.

Now, slope of the tangent of the function $p(x)$ at point $x = \frac{1}{2}$ is $p'(\frac{1}{2})$.

Hence $m = p'(\frac{1}{2})$.

So $m - \frac{3}{4} = 41$

Hence answer is 41.